

NFC-Tag-Specification

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Hardware

Mifare Ultralight C, as physical tag or emulated

Memory Map

The tag has 48 pages. Each page has 4 bytes. Pages 0 and 1 contain the serial number, Page 2 contains the last byte of the serial number, internal data, and two lock bytes. Page 3 contains one-time programmable data. Pages 4 to 39 contain user data, in total 144 bytes. Page 40 contains more lock bytes. Page 41 contains a counter. Pages 42 and 43 contain authentication configuration, and pages 44 to 47 contain the authentication key.

Data Protection

Tags are left at their factory default value, with all 144 user data bytes readable and writeable.

Data Encoding

Data is stored in the user data pages, as a list of records, padded by 0x00 bytes.

Record Format

Due to the very limited amount of memory on the tag, records are stored in a size-optimized binary format. Each record starts with a key byte, followed by binary data. The key byte is composed of type bits identifying the type of the binary data, and ID bits identifying the information in the binary data.

The ID portion of the key byte must be unique, even if the type bits change.

The binary can contain different data types, they are encoded by the most significant bits in the key byte:

Bit 7	Bit 6	Values	Data Format	Data Size, in bytes
0	0	0x00	If all other Bits are 0, final record. No data.	0
		0x01 to 0x3F	If at least one other Bit is 1, a NUL-terminated string, as in C.	At least 1, for the trailing NUL byte
0	1	0x40 to 0x7F	Binary IPv4 address, first byte of dotted decimal notation first.	4
1	0	0x80 to 0xBF	A single byte, used for booleans (0 = false, 1 = true), for flag bits, and for small numbers.	1
1	1	0xC0 to 0xFF	reserved for future use	will be defined later

Records

Category	Name	Required	Key ID (LSBs of the key byte)	Value type (MSBs of the key byte)	Key byte	Default	Description
Metadata	End of record list	yes	0x00	0x00 = end	0x00	N/A	End of record list
	Tag Flags	no	0x01	0x80 = byte	0x81	0x00	Tag Flag bits, see below
WiFi	SSID	yes	0x02	0x00 = string	0x02	factory default	WiFi name (SSID)
	Enterprise Identity	no	0x03	0x00 = string	0x03	""	WPA2 Enterprise identity. Ignored unless using WPA2 Enterprise.
	Enterprise Username	no	0x04	0x00 = string	0x04	""	WPA2 Enterprise username. Ignored unless using WPA2 Enterprise.
	Password	no	0x05	0x00 = string	0x05	""	WiFi password / WPA2 Enterprise password
	Enterprise mode	no	0x06	0x80 = byte	0x86	0xFF	WPA2 Enterprise authentication mode 0xFF: disable WPA2 Enterprise 0x00: EAP-TLS 0x01: EAP-PEAP 0x02: EAP-TTLS all others: reserved for future use
MQTT	Host	no	0x10	0x00 = string or 0x40 = IPv4 address	0x10 or 0x50	factory default	Hostname or IP address of MQTT server
	Username	no	0x11	0x00 = string	0x11	factory default	MQTT username
	Password	no	0x12	0x00 = string	0x12	factory default	MQTT password
IP	DHCPv4	no	-	-(boolean)	-	true	IPv4 DHCP mode Enabled if set to true. Set to false by setting an IPv4 address, see below.

Category	Name	Required	Key ID (LSBs of the key byte)	Value type (MSBs of the key byte)	Key byte	Default	Description
							Note: IPv6 addresses can not be set manually, this is a hardware limitation.
	IPv4 address	no	0x20	0x40 = IPv4 address	0x60	set by DHCP	IPv4 address, in dotted decimal form, e.g. "192.168.1.101" . Typically omitted. If specified, DHCP is disabled.
	IPv4 netmask / CIDR suffix	no	0x21	0x80 = byte	0xA1	derived from IPv4 address	Network mask / CIDR suffix , in decimal, i.e. the number of leading 1 bits in the network mask. 24 is equivalent to a network mask of 255.255.255.0. Typically omitted. Default is 8 if the first byte of the IP address is less than 128 (class A), 16 if the first byte of the IP address is between 128 and 191 (class B), else 24 (classes C, D, E). Ignored if DHCP is enabled.
	IPv4 default gateway	no	0x22	0x40 = IPv4 address	0x62	derived from IPv4 address and netmask	IPv4 default gateway, in dotted decimal form. Default is the IPv4 address bitwise AND the IPv4 network mask, then bitwise OR 0.0.0.1. This results in 10.0.0.1 for a 10.0.0.0/8 network, 192.168.42.1 for a 192.168.42.0/24 network. Ignored if DHCP is enabled.

Category	Name	Required	Key ID (LSBs of the key byte)	Value type (MSBs of the key byte)	Key byte	Default	Description
	First DNS server	no	0x23	0x40 = IPv4 address	0x63	default gateway	IP address of first DNS server. Ignored if DHCP is enabled.
	Second DNS server	no	0x24	0x40 = IPv4 address	0x64	0.0.0.0	IP address of second DNS server. Disabled if set to 0.0.0.0. Ignored if DHCP is enabled.
	Third DNS server	no	0x25	0x40 = IPv4 address	0x65	0.0.0.0	IP address of third DNS server. Disabled if set to 0.0.0.0. Ignored if DHCP is enabled.
SNTP	First SNTP server	no	0x30	0x00 = string or 0x40 = IPv4 address	0x30 or 0x70	"0.pool.ntp.org"	Hostname or IP address of first SNTP server Typically omitted
	Second SNTP server	no	0x31	0x00 = string or 0x40 = IPv4 address	0x31 or 0x71	"1.pool.ntp.org"	Hostname or IP address of second SNTP server Typically omitted
	Third SNTP server	no	0x32	0x00 = string or 0x40 = IPv4 address	0x32 or 0x72	"2.pool.ntp.org"	Hostname or IP address of third SNTP server Typically omitted
Future use	-	-	all others	all	all others		Reserved for future use

Tag Flag Bits

Bit	Name	Description
0	More tags	Set if data is continued on another tag (set on all but the last tag)
1	Incremental Update	Set to keep the existing data (must be set on continuation tags)
all other		reserved for future use, should be 0

Record Order

- Tag Flags (key ID 0x01) must be first record, if used.
- End of record list (key ID 0x00) must be last record, must always be present.
- Values in the "IP" category (key IDs 0x09 to 0x0E) must be sorted by ascending key ID.
- No specific order is required for other records. Sorted by key ID is recommended.

Examples

Note: All examples are untested!

Simple WiFi provided by common SOHO WiFi router

Using DHCP and defaults for almost everything, setting just the SSID and WiFi password

Bytes	Type	ID	Remarks
0x02 CheapRouter	0x00	String	WiFi SSID
0x05 t0p-\$ecr3t	0x00	String	WiFi Password
0x00	End of record list		Note: This 0x00 byte is not an end of string marker

Enterprise WiFi

Using DHCP and defaults for almost everything, setting just the SSID, WiFi username, and WiFi password

Bytes	Type	ID	Remarks
0x02 ExpensiveRouter	0x00	String	WiFi SSID
0x03 user0815@example.com	0x00	String	WiFi Identity
0x04 user0815	0x00	String	WiFi Username
0x05 t0p-\$ecr3t	0x00	String	WiFi Password
0x86 0x02	Byte	Enterprise Mode	EAP-TTLS
0x00	End of record list		Note: This 0x00 byte is not an end of string marker

Simple WiFi provided by common SOHO WiFi router, but no DHCP

Using defaults for almost everything, setting SSID, WiFi password, and IPv4 address (192.168.42.43). Netmask is derived from the IPv4 address (192 results in /24 = 255.255.255.0), default gateway from IPv4 address and netmask (192.168.42.1), DNS servers are set to the default gateway.

Bytes	Type	ID	Remarks
0x02 CheapRouter	0x00	String	WiFi SSID
0x05 t0p-\$ecr3t	0x00	String	WiFi Password
0x60 0xC0 0xA8 0x2A 0x2B	IPv4 address	IPv4 address of the RIoT	192.168.42.43
0x00	End of record list		

Cursed Enterprise WiFi without DHCP and very non-standard settings

Netmask 255.255.0.0 = /16, IP 10.6.78.90, Gateway 10.6.6.6, DNS servers 10.1.22.33, 10.22.33.1, and 10.33.1.22, NTP servers "time-01.pub.cursed.evil", "time-03.pub.cursed.evil", "time-04.pub.cursed.evil", MQTT server "riot-srv-01.dmz-02.cursed.evil".

Bytes					Type	ID	Remarks
0x02	CursedWiFi			0x00	String	WiFi SSID	
0x03	user0815@cursed.evil			0x00	String	WiFi Identity	
0x04	user0815			0x00	String	WiFi Username	
0x05	t0p-\$ecr3t			0x00	String	WiFi Password	
0x86	0x02				Byte	Enterprise Mode	EAP-TTLS
0x10	riot-srv-01.dmz-02.cursed.evil			0x00	String	MQTT server	
0x60	0x0A	0x06	0x4E	0x5A	IPv4 address	IPv4 address of the RIoT	10.6.78.90
0xA1	0x10				Byte	Netmask	/16 = 255.255.0.0
0x62	0x0A	0x06	0x06	0x06	IPv4 address	IPv4 address of the default gateway	10.6.6.6
0x63	0x0A	0x01	0x16	0x21	IPv4 address	IPv4 address of the first DNS server	10.1.22.33
0x64	0x0A	0x16	0x21	0x01	IPv4 address	IPv4 address of the second DNS server	10.22.33.1
0x65	0x0A	0x21	0x01	0x16	IPv4 address	IPv4 address of the third DNS server	10.33.1.22
0x30	time-01.pub.cursed.evil			0x00	String	First NTP server	
0x31	time-03.pub.cursed.evil			0x00	String	Second NTP server	
0x32	time-04.pub.cursed.evil			0x00	String	Third NTP server	
0x00					End of record list		

Cursed WiFi, split over three tags

Same as before, but device-specific settings on the first tag, and common settings on the other two tags.

Tag 1:

Bytes					Type	ID	Remarks
0x01	0x01				Byte	Flag bits	First tag Bit 0 set: more tags Bit 1 cleared: clear existing data
0x03	user0815@cursed.evil			0x00	String	WiFi Identity	
0x04	user0815			0x00	String	WiFi Username	
0x05	t0p-\$ecr3t			0x00	String	WiFi Password	
0x60	0x0A	0x06	0x4E	0x5A	IPv4 address	IPv4 address of the RIoT	10.6.78.90
0x00					End of record list		

Tag 2:

Bytes					Type	ID	Remarks
0x01	0x03				Byte	Flag bits	Middle tag Bit 0 set: more tags Bit 1 set: keep existing data
0x02	CursedWiFi		0x00		String	WiFi SSID	
0x86	0x02				Byte	Enterprise Mode	EAP-TTLS
0x10	riot-srv-01.dmz-02.cursed.evill		0x00		String	MQTT server	
0xA1	0x10				Byte	Netmask	/16 = 255.255.0.0
0x62	0x0A	0x06	0x06	0x06	IPv4 address	IPv4 address of the default gateway	10.6.6.6
0x63	0x0A	0x01	0x16	0x21	IPv4 address	IPv4 address of the first DNS server	10.1.22.33
0x64	0x0A	0x16	0x21	0x01	IPv4 address	IPv4 address of the second DNS server	10.22.33.1
0x65	0x0A	0x21	0x01	0x16	IPv4 address	IPv4 address of the third DNS server	10.33.1.22
0x00					End of record list		

Tag 3:

Bytes			Type	ID	Remarks
0x01	0x02		Byte	Flag bits	Last tag Bit 0 cleared: no more tags Bit 1 set: keep existing data
0x30	time-01.pub.cursed.evill	0x00	String	First NTP server	
0x31	time-03.pub.cursed.evill	0x00	String	Second NTP server	
0x32	time-04.pub.cursed.evill	0x00	String	Third NTP server	
0x00			End of record list		

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